

# mBGT: Encoding Brain Signals With Multimodal Brain Graph Transformer

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**Abstract**—Leveraging the multimodal brain signals collected from various electronic devices, such as electroencephalography (EEG) and functional near-infrared spectroscopy (fNIRS) data, has been regarded as a promising technique for automated brain disease diagnosis. Existing studies on multimodal brain signal analysis mainly focus on data alignment and basic feature integration, yet fail to capture the spatiotemporal correlations amongst diverse signals. This limitation results in the learning of biased brain signal representations. To eliminate this limitation, we propose a novel multimodal Brain Graph Transformer model, named mBGT. By learning both temporal and spatial dynamics exhibited in multiple brain signals, mBGT derives high-quality multimodal representations of brain signals from heterogeneous data sources. Specifically, we construct a dynamic brain graph based on simultaneous EEG-fNIRS data, which effectively embeds the spatiotemporal correlations amongst brain signals. Thereafter, we propose a spatiotemporal information enhanced graph Transformer to encode the constructed brain graph, extensively exploring the functional interactions among different brain regions. Extensive experiments are conducted on a public dataset to demonstrate the advantages of mBGT in terms of the performance in downstream detection tasks, compared to multiple state-of-the-art baselines. These highlight the superiority of mBGT in learning effective representations for brain graphs from multimodal brain signals.

**Index Terms**—Multimodal data fusion, brain disease diagnosis, graph Transformer, brain signals.

## I. INTRODUCTION

MULTIMODAL diagnosis, aimed at achieving effective disease diagnosis by fusing diverse information from different sources, stands as a prominent trend in the medical field nowadays [1], [2]. Particularly, with the development

of consumer electronics in neuroimaging, such as electroencephalography (EEG), functional magnetic resonance imaging (fMRI), and functional near-infrared spectroscopy (fNIRS), multimodal brain analysis has emerged as a pivotal approach for brain disease diagnosis [3], [4], where multiple electronic devices are deployed to monitor brain signals. Since the signals collected by different devices usually contain complementary information, fusing these signals to achieve reciprocal augmentation amongst them has been regarded as a promising solution to realize more accurate understanding of the brain; and ultimately, leading to more effective clinical diagnoses.

Recently, multimodal brain analysis via fusing two specific types of brain signals, EEG and fNIRS, has attracted extensive attention in both academic and industrial communities [5]. Both EEG and fNIRS rely on sensors (e.g., electrodes and optodes) that are placed on the brain scalp to monitor and collect brain signals from different brain regions. EEG signals record the neuronal electrical activities of brain regions over time, while fNIRS signals record the changes in blood oxygenation caused by neuronal activities [6]. These two types of signals have their own strengths and weaknesses. EEG usually exhibits superior temporal resolution at the millisecond-level, enabling the capture of rapid dynamic change of neuronal activity, and hence, has become one of the most popular tools for brain analysis [7], [8]. However, EEG has poor spatial resolution due to the limited propagation of electrical signals through the scalp and skull, which may hinder effective capture of spatial relations between brain regions. Conversely, fNIRS has high spatial resolution and can more effectively capture the spatial relations between different brain regions, despite its relatively low temporal resolution compared to EEG due to the delay in hemodynamic response. To leverage the complementary strengths of the EEG and fNIRS signals and to combine the best of two worlds, multimodal learning via the integration of the two signal is considered as a promising approach to further push the accuracy in brain analysis [9].

Many existing studies adopt deep learning for EEG-fNIRS signal fusion, due to its demonstrated effectiveness in extracting key features from multimodal data [10]. To enable data fusion, deep learning models are employed to extract representative features from different modalities, which are then seamlessly integrated into a unified representation. Despite the potential of deep learning-guided EEG-fNIRS fusion, a crucial challenge in the fusion of the two signals is to resolve their disparities in temporal and spatial resolutions [11]. Existing works treat this as a feature alignment task, that is, simply

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adjusting and mapping these two signals onto a consistent feature space. However, such an approach may lead to representation biases, since it only addresses the disparities of two different signals but overlooks the inherent spatiotemporal relations between two modalities. To make the best use of the strengths offered by each modality and obtain high-quality multimodal representations, we argue that it is crucial to capture the intricate spatiotemporal relations between the EEG and fNIRS signals when fusing them.

To address this issue, this paper proposes a novel **multimodal Brain Graph Transformer** model, namely mBGT, which leverages graph learning to model the spatiotemporal relations of EEG-fNIRS signals, obtaining high-quality multimodal representations of EEG-fNIRS signals. Notably, graph structure has become a primary form to present the complex functional correlations and structural connections within the brain, forming brain graphs (also known as brain networks) [12]. In addition, brain graphs constructed based on brain signals can depict the correlations between these signals, revealing the interaction patterns of signals, as well as the functional connectivity within the brain [7]. Therefore, our proposed mBGT is based on graph learning, specifically the graph Transformer, modeling the spatiotemporal relations of EEG-fNIRS signals and achieving high-quality multimodal brain graph representation.

In mBGT, we first apply a linear interpolation-based data alignment method on EEG and fNIRS signals to ensure that they reflect the same brain activity at the same timestamp. Subsequently, a dynamic brain graph is meticulously constructed based on the multimodal information derived from the aligned EEG-fNIRS data, where each node in the graph corresponds to a signal channel, i.e., a region of interest (ROI), and the edges represent the correlations between different brain regions. We build a sequence of such graphs, where at each graph, the original temporal information of EEG signals is integrated with fNIRS signals, forming node features and fully leveraging the advantages of EEG's high temporal resolution. Furthermore, the edges in each graph are constructed based on the correlations among fNIRS signals, which possess high spatial resolution. The constructed dynamic brain graph can effectively capture the spatiotemporal relation of multimodal brain signals, obtaining enriched information in brain signal data. Finally, a spatiotemporal information enhanced graph Transformer model is applied over the constructed dynamic brain graph to capture the multidimensional correlations amongst signals, which ultimately, derive effective representations for brain signals.

mBGT can learn high-quality multimodal representations of EEG-fNIRS data, fully utilizing the strengths offered by each modality to explore the functional interactions among brain regions. Our experiments show that mBGT generates more effective representations of brain graphs, exploring the signal transmission mechanisms among brain regions, and further, improves the performance of key medical tasks such as brain disease diagnosis. Particularly, mBGT shows great potential in improving the brain-computer interface (BCI) techniques with the ability in accurate multimodal brain signals analysis. Moreover, because of its superiority in sufficient information

extraction, our method can be applied to assist the development of efficient neurological disorder treatment. The main contributions of this paper are as follows:

- 1) We propose to construct a dynamic brain graph based on the EEG-fNIRS data, capitalizing on the high temporal resolution of EEG and the superior spatial resolution of fNIRS.
- 2) We propose a multimodal Brain Graph Transformer model, named mBGT, for learning high-quality representations of multimodal brain signals, effectively capturing their inherent spatiotemporal relations.
- 3) We conduct extensive experiments on a public dataset, demonstrating the effectiveness of the proposed model over current state-of-the-art methods.

The rest of the paper is organized as follows. Section II reviews related work. Section III revisits some principal information on brain graph and Transformer. Section IV introduces on the proposed mBGT. Section V presents experimental setting and results. Finally, we conclude our work in Section VI.

## II. RELATED WORK

In this section, we review relevant literature to this paper, covering (1) multimodal medical data mining, (2) brain graph analysis, and (3) graph Transformer.

### A. Multimodal Medical Data Mining

With the development of medical consumer electronics, medical health data in various modalities can be obtained such as electronic health records (EHRs), molecular data, digital pathology, and medical images. These various types of data record clinical procedures such as patient visits and lab tests rigorously, and hence, lays a foundation for a broad range of digital health applications, including digital clinical trials, individualized medicine and virtual health coaches [13], [14]. In particular, multimodal diagnosis enabled by the integration of different data modalities collected from various sources has attracted extensive attention, due to its ability to capture the complexity of disease and the human health condition in general [15], [16], [17]. Particularly, combining multimodal medical data benefits clinical analysis, ultimately providing decision-making support for both patients and medical establishments [18].

Many multimodal data fusion approaches have been proposed for the purpose of integrating medical data [19], [20], [21]. *M<sup>3</sup>Care* [22] considers EHR as multimodal data comprised of structured components of different modalities, such as diagnoses, laboratory values, demographic information, and vital signs. *M<sup>3</sup>Care* combines the multimodal healthcare data in EHR to perform clinical analysis. Steyaert et al. [23] proposed a data fusion strategy, integrating multimodal biomarker data (e.g., EHR, molecular data, radiology and pathology) to address data sparsity, data orchestration and model interpretability issues for cancer discovery. Muhammad et al. [24] presented a review of recent advancements in fusing medical signals — extracting and integrating abundant features from different types

of physiological signals, like electrocardiogram (ECG), electroglottography (EGG), electroencephalogram (EEG), and electrooculogram (EOG).

Specifically, multimodal brain health data fusion has attracted great attention in the area of brain disease diagnosis and analysis [25], [26]. MMGL [27] aggregates the features of functional Magnetic Resonance Imaging (fMRI) data and Diffusion Tensor Imaging (DTI) data by leveraging the correlation between two modalities, exploiting richer information associated with brain diseases. HetMed [28] focuses on multimodal medical image analysis, fusing image and non-image medical data based on a graph structure. More recently, several works have focused on the multimodal EEG-fNIRS analysis, which utilizes the complementary information carried by EEG and fNIRS signals to achieve better clinical diagnostic efficacy [5], [10]. However, most of current works simply align the features of EEG and fNIRS signals to achieve multimodal fusion, ignoring the inherent spatiotemporal correlations between the aligned signals, leading to representation biases. In this paper, we study the multimodal EEG-fNIRS data fusion for encoding brain networks with a specific focus on exploring their spatiotemporal correlations. Therefore, our model can learn high-quality multimodal brain signal representations, which contain more sufficient information. Furthermore, our method can fully utilize the strengths of EEG and fNIRS to effectively explore the functional interactions among brain regions.

### B. Brain Graph Analysis

Brain Graphs, reflecting the connections in human neural system, can be constructed from various types of brain signals, such as functional magnetic resonance imaging (fMRI), positron emission tomography (PET) and electroencephalography (EEG) [12], [29]. Generally, according to anatomical parcellation schemes, brain graphs are undirected graphs with regions of interest (ROIs) as nodes and the connectivities between ROIs represented as edges [7], [30], [31]. Notably, the undirected nature of brain graphs indicates that there is no direction of the information flow between regions. Brain graphs can be divided into three types in terms of different analysis perspectives, including functional brain graphs, structural brain graphs and morphological brain graphs [12]. Understanding brain graphs is of great significance, as it facilitates us to understand the underpinning functional and structural patterns in the brain. Therefore, recently, brain graph analysis has attracted increased attention in both the domain of computer science and neuroscience.

More recently, graph learning-based brain graph analysis has been more and more popular, especially in the domain of bioinformatics [32], [33], [34]. BrainPrint [32] proposes a network estimation module and a graph analysis module to extract features for EEG biometric identification. BrainGNN [35] is a graph neural network model for fMRI analysis, containing special ROI-aware graph convolutional layers that leverage the functional information of brain networks. BrainGB [30] summarizes the pipelines of brain graph construction and analyzes the feasibility of various graph

neural network models for analyzing brain graphs. Compared to studies working on static brain graphs, some works focus on exploring dynamic brain graphs, that is, capturing the temporal changes in brain graphs. For example, STAGIN [36] uses a spatio-temporal attention mechanism in graph neural networks (GNNs) to learn the dynamic brain graph representation. GGN [31] utilizes a multi-layer temporal encoder and a spatial decoder in the functional components to discover the dynamic brain functional connectivity based on scalp EEG signals. In this paper, we capture the temporal changes in brain graphs by building a dynamic brain graph due to its ability in conveying comprehensive information over time. Especially, we construct the dynamic brain graph based on multimodal brain signal data, fully taking advantage of the strengths of each modality and obtaining enriched temporal and spatial information extracted from brain signals.

### C. Graph Transformer

With Transformer showing great superiority in both natural language processing (NLP) [37] and computer vision (CV) [38], [39], many recent works attempt to generalize Transformer models to graph data [40], [41]. Graph Transformer [42], one of the most popular model, aims to encode graph topological features by defining an attention mechanism for edge embedding and represents positional encoding by the Laplacian eigenvectors. With such a structure, two common challenges in graph learning can be alleviated, that is, effectively encoding structural information and capturing the long-term dependencies between nodes [43], [44]. It then has various applications in graph learning. HGT [45] applies graph Transformer to larger-scale heterogeneous graphs. Graphomer [46] proposes three structural encoding methods, including centrality encoding, spatial encoding and edge encoding, to model network-structured data. ANS-GT [47] utilizes a multi-armed bandit algorithm to achieve adaptive node sampling and long-term dependency capture.

There are an increasing number of studies adapting graph Transformer models to explore biological networks (e.g., molecular graphs), with specific designs/modifications to the graph Transformer models tailored for biological networks. For example, GROVE [48] presents Transformer-based self-supervised message passing networks to learn rich semantic and structural information of molecular graphs. KPGT [49] introduces a Line Graph Transformer (LiGhT), containing path encoding and distance encoding, to embed the structural information of molecular graphs. DeepMAPS [50] provides a multi-head graph Transformer to learn the connectivities among cells as well as achieve biological network inference. Brain network Transformer [51] tries to develop a Transformer-based model to analyze brain networks. More recently, Brain TokenGT [52], a brain functional connectome trajectory embedding framework, leverages graph Transformer model to model spatiotemporal information in brain graphs. Because of the tremendous potential of graph Transformer in capturing the spatial information in brains, this work develops a spatiotemporal information enhanced graph Transformer to encode brain graphs.

TABLE I  
 NOTATIONS IN THIS PAPER

| Notations            | Description                                                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------------------|
| $\mathcal{V}$        | Node set                                                                                                           |
| $\mathcal{E}$        | Edge set                                                                                                           |
| $\mathcal{G}$        | Brain graph, $\mathcal{G} = (\mathcal{V}, \mathcal{E})$                                                            |
| $v_i$                | The $i$ -th node (the $i$ -th ROI) in the brain graph                                                              |
| $e_{ij}$             | Edge between node $v_i$ and $v_j$                                                                                  |
| $n$                  | Number of nodes                                                                                                    |
| $S_{e_{ij}}$         | Strength of edge, $S(e_{ij}) > 0$                                                                                  |
| $\mathbf{H}$         | Node feature matrix, $\mathbf{H} \in \mathbb{R}^{n \times d}$                                                      |
| $\mathbf{h}_i$       | Feature vector of node $v_i$ , $\mathbf{h}_i \in \mathbb{R}^d$                                                     |
| $G$                  | Brain graph sequence in the time series of fNIRS                                                                   |
| $\mathcal{G}_t$      | Brain graph at time $t$ , $\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t)$                                         |
| $\mathbf{Q}^{l,m}$   | Query representation of the $m$ -th self-attention head at layer $l$                                               |
| $\mathbf{K}^{l,m}$   | Key representation of the $m$ -th self-attention head at layer $l$                                                 |
| $\mathbf{V}^{l,m}$   | Value representation of the $m$ -th self-attention head at layer $l$                                               |
| $d$                  | Dimension of node feature embeddings                                                                               |
| $d_m$                | Dimension of each self-attention head                                                                              |
| $\mathbf{W}_Q^{l,m}$ | Projection matrices of the $m$ -th self-attention head at layer $l$ for query representation                       |
| $\mathbf{W}_K^{l,m}$ | Projection matrices of the $m$ -th self-attention head at layer $l$ for key representation                         |
| $\mathbf{W}_V^{l,m}$ | Projection matrices of the $m$ -th self-attention head at layer $l$ for value representation                       |
| $\mathbf{A}^{l,m}$   | Attention matrix capturing the similarity between queries and keys of the $m$ -th self-attention head at layer $l$ |
| $\mathbf{A}_t^s$     | Spatial attention matrix at time $t$                                                                               |
| $M$                  | Number of self-attention heads                                                                                     |
| $T_f$                | Time series of fNIRS data                                                                                          |
| $T_e$                | Time series of EEG data                                                                                            |
| $q_f$                | Sampling frequencies of fNIRS                                                                                      |
| $q_e$                | Sampling frequencies of EEG                                                                                        |
| $F_t^i$              | fNIRS value of node $i$ at time $t$                                                                                |
| $E_t^i$              | EEG value of node $i$ at time $t$                                                                                  |

### III. PRELIMINARIES

In this section, we recap the definition of brain graphs and the Transformer model. The notations that are frequently used in this paper are summarized in Table I.

**Brain Graphs.** A brain graph is the brain functional connectome (FC) with ROIs (signal channels) as the nodes and the correlations among nodes as the edges. It is denoted as  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ , where  $\mathcal{V} = \{v_1, v_2, \dots, v_n\}$  is the node set and  $\mathcal{E}$  means the set of edges,  $n$  is the number of nodes.  $e_{ij} \in \mathcal{E}$  stands for the edge between nodes  $v_i \in \mathcal{V}$  and  $v_j \in \mathcal{V}$ . The brain graph is an undirected weighted graph. According to degrees of the connectivities between ROIs, each edge assigned with a weight, denoted as  $S_{e_{ij}} > 0$ . When  $S_{e_{ij}} = 0$ , there is no edge between  $v_i$  and  $v_j$ . We define the node feature matrix as  $\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_n]^T$ , here  $\mathbf{h}_i$  indicates the feature vector of node  $v_i$ .

In the time series of fNIRS, a sequence of brain graphs are represented as  $G = (\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_T)$ , where  $\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t)$  indicates the brain graph at time  $t$ . It should be noted that although the number of nodes remains constant in the brain graphs captured at different time stamps, the number of edges and their corresponding weights may vary. The features of nodes at different timestamps may vary too.

**Transformer.** A classic Transformer architecture is composed of multiple Transformer layers, each of which includes a multi-head self-attention module and a feed-forward network

(FFN). Here, we briefly present the computation of the multi-head self-attention module. Given the feature matrix  $\mathbf{H} \in \mathbb{R}^{n \times d}$ , where  $d$  denotes the hidden dimension, the query representation  $\mathbf{Q}^{l,m}$ , key representation  $\mathbf{K}^{l,m}$  and value representation  $\mathbf{V}^{l,m}$  of the  $m$ -th self-attention head at layer  $l$  are calculated as:

$$\begin{aligned} \mathbf{Q}^{l,m} &= \mathbf{H}^{l-1} \mathbf{W}_Q^{l,m}, \\ \mathbf{K}^{l,m} &= \mathbf{H}^{l-1} \mathbf{W}_K^{l,m}, \\ \mathbf{V}^{l,m} &= \mathbf{H}^{l-1} \mathbf{W}_V^{l,m}, \end{aligned} \quad (1)$$

where  $\mathbf{W}_Q^{l,m} \in \mathbb{R}^{d \times d_m}$ ,  $\mathbf{W}_K^{l,m} \in \mathbb{R}^{d \times d_m}$  and  $\mathbf{W}_V^{l,m} \in \mathbb{R}^{d \times d_m}$  are the projection matrices of the  $m$ -th self-attention head at layer  $l$ .  $d_m = \frac{d}{M}$  is the dimension of each head, where  $M$  means the number of heads.  $\mathbf{H}^{l-1}$  indicates the feature matrix at the  $(l-1)$ -th layer. Then, the similarity between queries and keys can be captured:

$$\mathbf{A}^{l,m} = \frac{\mathbf{Q}^{l,m} (\mathbf{K}^{l,m})^T}{\sqrt{d_m}}. \quad (2)$$

Afterwards,  $\mathbf{H}^l$ , the feature matrix at layer  $l$ , can be obtained:

$$\begin{aligned} \mathbf{H}^{l,m} &= \text{softmax}(\mathbf{A}^{l,m}) \mathbf{V}^{l,m}, \\ \mathbf{H}^l &= \text{concat}(\mathbf{H}^{l,1}, \mathbf{H}^{l,2}, \dots, \mathbf{H}^{l,M}). \end{aligned} \quad (3)$$

Here,  $\text{softmax}(\cdot)$  and  $\text{concat}(\cdot)$  are the softmax and concatenation operation, respectively.

### IV. THE PROPOSED METHOD

In this section, we introduce the proposed mBGT in detail. In general, mBGT contains three main modules, including the multimodal data alignment, dynamic brain graph construction and spatiotemporal information enhanced graph Transformer module. Fig. 1 shows the overall framework of mBGT. Firstly, we align the EEG and fNIRS data with different time resolution onto the same time series. Then, a dynamic brain graph is constructed based on the paired EEG-fNIRS data, extracting the spatiotemporal relations of multimodal data. Lastly, we apply a spatiotemporal information enhanced graph Transformer model on the fused data to learn an effective representations of brain graphs.

#### A. Multimodal Data Alignment

Due to the disparity in terms of temporal resolution of the EEG and fNIRS data, the two types of signals must be aligned onto a same set of timesteps before feature fusion. Thus, we apply an alignment approach based on the linear interpolation to preprocess the data. Fig. 2 illustrates the data alignment procedure.

As shown in Fig. 2, the time resolution of fNIRS data is relatively low, while that of EEG data is extremely high. The set of timestamps of fNIRS data is denoted as  $T_f = (t_f^1, t_f^2, \dots, t_f^k)$ , and the set of timestamps of EEG is denoted as  $T_e = (t_e^1, t_e^2, \dots, t_e^z)$ . Here,  $k$  and  $z$  are the number of the timestamps recording fNIRS and EEG data respectively, noting that  $k < z$ . The overall time periods of sampled fNIRS

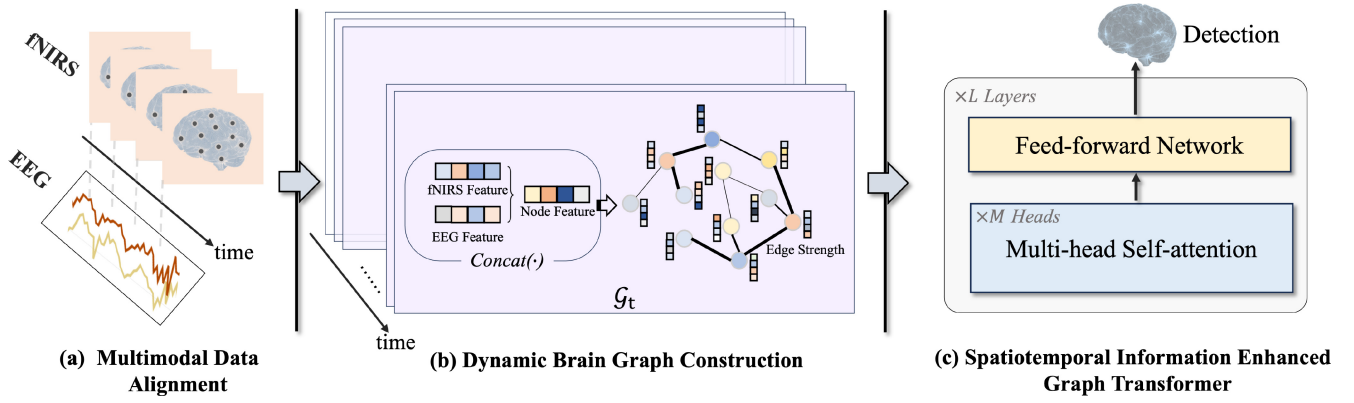


Fig. 1. The illustration of mBGT framework.

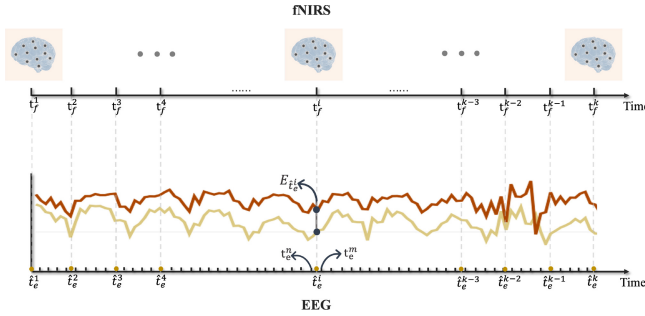


Fig. 2. An illustration of multimodal signal alignment.

and EEG data are the same, while EEG has higher sampling frequency. We denote the sampling frequencies of fNIRS and EEG data as  $q_f$  and  $q_e$ , with  $q_f < q_e$ . Then, we adjust the EEG data using the timestamps of fNIRS data:

$$\begin{aligned} \hat{T}_e &= \{\hat{t}_e^1, \hat{t}_e^2, \dots, \hat{t}_e^k\} \\ &= \left\{ \frac{t_f^i \times q_e}{q_f} \mid i = 1, 2, \dots, k \right\}. \end{aligned} \quad (4)$$

Here,  $\hat{T}_e = \{\hat{t}_e^1, \hat{t}_e^2, \dots, \hat{t}_e^k\}$  represents the timestamps to be used for EEG data after alignment. To further ensure that the aligned timestamps of EEG data is consistent with those of fNIRS, we utilize the least squares to optimize the alignment result. The formula is as follows:

$$\text{minimize} \sum_i (t_f^i - \hat{t}_e^i)^2. \quad (5)$$

Then, we estimate the value of EEG at each timestamp in  $\hat{T}_e = \{\hat{t}_e^1, \hat{t}_e^2, \dots, \hat{t}_e^k\}$ . It is worth noting that the EEG value at a certain timestamp refers to the electrical activity information of neurons at this time point. As shown in Fig. 2, the corresponding EEG data at each timestamp in  $T_e = (t_e^1, t_e^2, \dots, t_e^k)$  is sampled. However, the EEG values in series of timestamps  $\hat{T}_e = \{\hat{t}_e^1, \hat{t}_e^2, \dots, \hat{t}_e^k\}$  may be absent in the original EEG data series. Therefore, we need to compute those missing EEG values for  $\hat{T}_e = \{\hat{t}_e^1, \hat{t}_e^2, \dots, \hat{t}_e^k\}$ . Especially, we estimate the EEG values utilizing linear interpolation. Firstly, for time  $\hat{t}_e^i$ , we find two timestamps in  $T_e = (t_e^1, t_e^2, \dots, t_e^k)$  that are closest to it. Assuming the two identified timestamps are

$t_e^n$  and  $t_e^m$ , where  $t_e^n < \hat{t}_e^i < t_e^m$ . Then, the EEG values at time  $t_e^n$  and  $t_e^m$  are obtained by integrating the recorded electrical activity information of all the neurons, denoted as  $E_{t_e^n}$  and  $E_{t_e^m}$ , respectively. Afterwards, the EEG value at time  $\hat{t}_e^i$  can be computed using linear interpolation:

$$E_{\hat{t}_e^i} = E_{t_e^n} + \frac{E_{t_e^m} - E_{t_e^n}}{t_e^m - t_e^n} \times (\hat{t}_e^i - t_e^n). \quad (6)$$

## B. Dynamic Brain Graph Construction

We construct a dynamic brain graph based on the aligned multimodal data, focusing on capturing the spatiotemporal relations of brain signals and revealing the signal transmission mechanisms. The spatial and temporal information conveyed by the constructed dynamic brain graph can fully reflect the brain functional connectivities. Specifically, we build the dynamic brain graph that is composed of a sequence of graphs, denoted as  $G = (G_1, G_2, \dots, G_k)$ . The nodes in each brain graph are the ROIs (signal channels of EEG and fNIRS data) and edges represent the correlation of nodes. Note that it is assumed that the EEG and fNIRS data have the signal channels corresponding to the same set of ROIs. The number of graphs is  $k$ , which is the number of fNIRS timestamps. That means one sampling of fNIRS corresponds to a brain graph. Particularly, we first combine the temporal information of EEG signals with fNIRS data and embed it into the node features, fully leveraging the advantages of EEG's high temporal resolution. Then, because of the high spatial resolution of fNIRS data, we calculate the weight of each edge based on the fNIRS signal, reflecting the functional correlation of nodes.

1) *Node Feature Extraction*: For  $G_t$ , to obtain the features of each node, we extract both the fNIRS and EEG information at time  $t$  and fuse them together. EEG data records the electrical activity of nodes with a high temporal resolution, while fNIRS shows the changes in blood oxygenation with larger time intervals. Fusing EEG temporal information with fNIRS data can effectively capture the dynamic changes of brain activity. Significantly, by embedding the information from aligned EEG-fNIRS data into a unified dynamic brain graph, we can capture the underlying spatiotemporal relations within brain signals. This approach also fully utilizes the

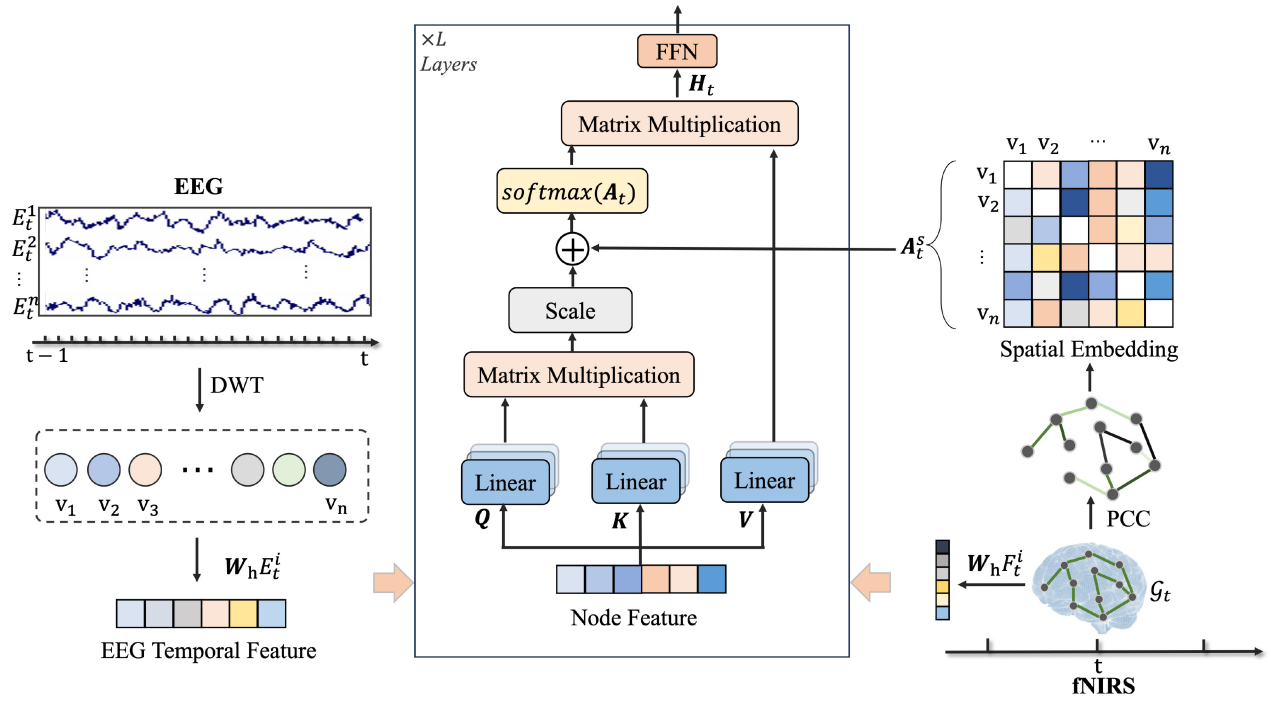


Fig. 3. The architecture of ST-Graph Transformer.

strengths of both EEG and fNIRS data, enhancing the quality of representations.

For node  $v_i$  in  $\mathcal{G}_t$ , we first calculate its fNIRS value at time  $t$ , denoted as  $F_t^i$ . Then, the EEG value  $E_t^i$  is obtained by integrating the electrical activity of nodes from time  $t-1$  to  $t$  via discrete wavelet transform (DWT). The equation is as follows:

$$E_t^i = C(a, b) = \sum_{t=1}^t E_t^i \times \psi\left(\frac{t-b}{a}\right). \quad (7)$$

Here,  $C(a, b)$  is the wavelet coefficient, of which the scale and shift are  $a$  and  $b$ , respectively.  $\psi(\cdot)$  is the wavelet function. Afterwards, we use concatenation operation to combine fNIRS feature  $F_t^i$  with EEG feature  $E_t^i$ :

$$\mathbf{h}_t^i = \text{concat}(\mathbf{W}_h F_t^i, \mathbf{W}_h E_t^i), \quad (8)$$

where  $\mathbf{W}_h$  indicates the projection matrix.  $\mathbf{h}_t^i$  is the node feature of  $i$  at time  $t$ . Consequently, the node feature matrix at timestamp  $t$  can be obtained as:

$$\mathbf{H}_t = [\mathbf{h}_1^t, \mathbf{h}_2^t, \dots, \mathbf{h}_n^t]^\top. \quad (9)$$

2) *Computation of Edge Weights*: A brain graph is a weighted graph with each edge being assigned a weight that represents the functional correlation of nodes. Because of the high spatial resolution of fNIRS signals, this paper computes the edge weights based on fNIRS data. Therefore, the functional interactions between brain regions can be effectively revealed, as well as delineating the spatial distribution of brain activity.

To obtain the edge weights, we first compute the Pearson correlation coefficient (PCC) of nodes according to their

fNIRS features. Hence, at time  $t$ , the PCC value of node  $v_i$  and  $v_j$  is as follows:

$$P_t(v_i, v_j) = \frac{\sigma(v_i, v_j)}{\delta(v_i) \times \delta(v_j)}. \quad (10)$$

Here,  $\sigma(v_i, v_j)$  is the covariance of  $v_i$  and  $v_j$ ,  $\delta(\cdot)$  denotes the standard deviations.

Then, we use an exponential function  $\text{EXP}(\cdot)$  to calculate the weight of the edge between node  $v_i$  and  $v_j$  at time  $t$ :

$$S_{e_{ij}}^t = \text{EXP}[P_t(v_i, v_j)]. \quad (11)$$

It is worth noting that the weight of an edge changes over time due to the changes in brain state and functions. Therefore, each graph in the dynamic brain graph  $\mathcal{G}$  is different in terms of the edge number and edge weights, as well as the node features.

### C. Spatiotemporal Information Enhanced Graph Transformer

In this section, we propose a novel spatiotemporal information enhanced graph Transformer architecture, called ST-Graph Transformer, to learn brain graph representations for downstream tasks. Fig. 3 illustrates the architecture of ST-Graph Transformer, which is built upon the vanilla Transformer [53]. For a dynamic brain graph  $\mathcal{G}$ , ST-Graph Transformer learns the graph of each timestamp.

Given the brain graph at time  $t$ , denoted as  $\mathcal{G}_t$ , we extract node feature matrix  $\mathbf{H}_t$  combining EEG and fNIRS feature as the input of ST-Graph Transformer. Based on Graphormer [46], we design a spatial embedding to encode its structural information into Transformer model (see Fig. 3). In particular, we leverage the weight of each edge to represent the spatial relations of nodes, where  $S_{e_{ij}}^t$  represents the edge

weight between nodes  $v_i$  and  $v_j$  at time  $t$ . Thereafter, a spatial attention matrix  $\mathbf{A}_t^s$  can be obtained by embedding  $S_{eij}^t$  into the latent feature space:

$$\begin{aligned}\mathbf{A}_t^s &= [\mathbf{a}_{11}^s, \mathbf{a}_{12}^s, \dots, \mathbf{a}_{nn}^s]^\top, \\ \mathbf{a}_{ij}^s &= \mathbf{W}_s S_{eij}^t.\end{aligned}\quad (12)$$

where  $\mathbf{a}_{ij}^s \in \mathbf{A}_t^s$  is the attention scalar,  $\mathbf{W}_s \in \mathbb{R}^{d_m}$  indicates the projection matrix. When  $i = j$ ,  $\mathbf{a}_{ij}^s = 0$ .

Afterwards, we recalculate the attention matrix  $\mathbf{A}^{l,m}$  in Equation 2 by injecting the spatial information. Specifically, we compute  $\mathbf{A}_t^{l,m}$  as follows:

$$\mathbf{A}_t^{l,m} = \frac{\mathbf{H}_t^{l-1} \mathbf{W}_Q^{l,m} (\mathbf{H}_t^{l-1} \mathbf{W}_K^{l,m})^\top}{\sqrt{d_m}} + \mathbf{A}_t^s. \quad (13)$$

Then, the feature matrix at layer  $l$  is obtained as:

$$\begin{aligned}\mathbf{H}_t^{l,m} &= \text{softmax}(\mathbf{A}_t^{l,m}) \mathbf{H}_t^{l-1} \mathbf{W}_V^{l,m}, \\ \mathbf{H}_t^l &= \text{concat}(\mathbf{H}_t^{l,1}, \mathbf{H}_t^{l,2}, \dots, \mathbf{H}_t^{l,M}).\end{aligned}\quad (14)$$

After the concatenation operation, the feature matrix  $\mathbf{H}_t^l$  is fed to a FFN module:

$$\begin{aligned}\tilde{\mathbf{H}}_t^l &= \text{LN}(\mathbf{H}_t^{l-1} + \mathbf{H}_t^l), \\ \mathbf{H}_t^l &= \text{LN}(\mathbf{W}_2^l \text{ReLU}(\mathbf{W}_1^l \tilde{\mathbf{H}}_t^l)) + \tilde{\mathbf{H}}_t^l.\end{aligned}\quad (15)$$

Here,  $\text{LN}(\cdot)$  is the Layer normalization operator,  $\text{ReLU}(\cdot)$  stands for the activation function.  $\mathbf{W}_2^l$  and  $\mathbf{W}_1^l$  are the projection matrices.

In conclusion, ST-Graph Transformer takes the node features, which combines the temporal EEG and fNIRS features, as input, capturing the temporal dynamics of brain graph. In addition, a novel edge strength-based spatial embedding module is developed for structural information encoding.

## V. EXPERIMENTS

In this section, we first evaluate our model in terms of its performance on two mental detection tasks, including mental arithmetic (MA) task and motor imagery (MI) task. These two tasks aim to demonstrate our model can accurately detect brain signals in different mental states, showing its potential in accurate brain pattern identification and brain disease diagnosis. Then, we conduct an ablation study to prove the effectiveness of each component of mBGT.

### A. Experimental Settings

1) *Datasets*: In this paper, a benchmark dataset [54], which simultaneously records EEG and fNIRS signals, is used to evaluate our model. This dataset totally contains 29 subjects (14 males and 15 females), of which 28 are right-hand subjects and 1 is a left-hand subject. The average age of all the subjects is  $28.5 \pm 3.7$  (mean  $\pm$  standard deviation). EEG signals are recorded at 1000-Hz sampling rate from 30 channels, while fNIRS signals are recorded at 12.5-Hz sampling rate from 36 channels consisting of 14 sources and 16 detectors. Afterwards, 30 EEG signals and 36 fNIRS

signals are downsampled to 200 Hz and 10 Hz, respectively. Therefore, after data alignment, we can totally extract 36 nodes to construct the brain graph according to fNIRS channels, each of which represents one ROI. It is worth noting that we match EEG and fNIRS channels based on the electrode positions, and then embed each matched channel pair into one node as its feature. For nodes that lack EEG information, we set their EEG features to null.

The dataset is divided into two subsets according to two tasks: mental arithmetic (MA) task and motor imagery (MI) task.

- *MA dataset*: In the MA task, all subjects are asked to repeatedly perform subtraction during the 10-second task period. After one task period, subjects can take a rest. In one trial, subjects repeat this test-rest cycle 20 times. Totally, there are 30 trials for each subject.
- *MI dataset*: For the MI task, subjects are asked to perform kinesthetic MI, imagining the opening and closing of their hands. Subjects are asked to do 30 trials, each of which includes 20 same sessions. Each session starts with a 2-second visual instruction, subjects then are required to imagine opening and closing their hands with a 1-Hz pace. Subjects should continuously perform the imagine task during the task period, which lasts 10 seconds. Then, a 2-second resting period, during which subjects rest without any thoughts, is provided after one task period. Among the 20 sessions, 10 sessions each for left-hand and right-hand MI.

2) *Evaluation Metrics*: In our experiment, the MI task and MA task are both detection tasks used to determine whether the subject's brain, at a certain time in the time series of the dynamic brain graph, is in a task state or a resting state. Thereby, the MI and MA tasks are both graph classification tasks, which are binary classification tasks. Generally, we choose accuracy, F1 score and ROC curve as the evaluation metrics.

Accuracy is the ratio of the graphs that are correctly classified out of all samples. Given  $S$  test samples,  $TP$ ,  $TN$ ,  $FP$  and  $FN$  are the counts of true positives, true negatives, false positives and false negatives, respectively. The formula of accuracy is as follows:

$$\begin{aligned}\text{Accuracy} &= \frac{TP + TN}{S} \\ &= \frac{TP + TN}{TP + TN + FP + FN}.\end{aligned}\quad (16)$$

In addition, F1, which the harmonic mean of precision and recall, is also calculated to evaluate our model. Firstly, the precision  $P$  and recall  $R$  are calculated:

$$\begin{aligned}P &= \frac{TP}{TP + FP}, \\ R &= \frac{TP}{TP + FN}.\end{aligned}\quad (17)$$

Then, F1 score is calculated:

$$F1 = \frac{2 \times P \times R}{P + R}.\quad (18)$$

For all of the above metrics, higher values indicate better performances.

In order to comprehensively evaluate our model, we also use the ROC (receiver operating characteristic) curve as an evaluation metric. The ROC curve shows the trade-off between true positive rate (TPR) and false positive rate (FPR) during the detection process. The formulas of TPR and FPR are as follows:

$$\begin{aligned} TPR &= \frac{TP}{TP + FN}, \\ FPR &= \frac{FP}{FP + TN}. \end{aligned} \quad (19)$$

One metric used to comprehensively evaluate the ROC curve is AUC (area under the curve), which represents the area under the ROC curve. The value of AUC ranges from 0 to 1. A higher AUC indicates a better model performance.

3) *Baselines*: To evaluate the effectiveness of the proposed mBGT, we compare it against the following 5 state-of-the-art works, which focus on deep learning-based EEG-fNIRS signal fusion:

- **Shrinkage LDA [54]** aims to combine EEG and fNIRS signals for brain status detection. Additionally, the author also releases an open EEG-fNIRS dataset, which is used in this paper. This paper performs basic decoding procedures on the released dataset to showcase the superiority of utilizing multimodal data for brain function analysis. The EEG and fNIRS features are simply integrated for downstream tasks. Particularly, a shrinkage linear discriminant analysis (LDA) is adapted to conduct experiments for classification tasks.
- **IDPF [55]** first adopts some independent decision paths, each of which analyzes different brain state information. Specifically, in each decision path, fNIRS feature is used to modify the EEG-based common spatial pattern algorithm. Then, these independent decision paths are integrated via a reliability-based decision-level fusion strategy.
- **Signal Fusion [56]** proposes three fusion methods, including linear fusion, tensor fusion and  $p$ th-order polynomial fusion, to integrate the EEG and fNIRS features. The linear fusion method aims to simply concatenate the feature vectors of two modalities, while tensor fusion method utilizes an outer product to combine the EEG and fNIRS features. In addition,  $p$ th-order polynomial fusion method uses both concatenation operation and outer product.
- **FGANet [11]** is a deep learning-based early feature fusion model, integrating EEG and fNIRS features before the fully-connected layer of convolutional neural network. Particularly, a fNIRS-guided attention layer is introduced to extract the joint representations of the EEG-fNIRS tensor.
- **GLM+CSP [9]** firstly proposes an EEG-informed fNIRS general linear model (GLM) to extract fNIRS features. Afterwards, combining the extracted fNIRS features with the common spatial pattern (CSP) features obtained from an optimal narrowband of EEG to gain EEG-fNIRS features.

Furthermore, we compare the performance of mBGT to other representative graph Transformer models, including

TABLE II  
THE CLASSIFICATION ACCURACY (MEAN±STANDARD DEVIATION) OF DIFFERENT MODELS IN MI AND MA TASK

| Method                              | MI task           | MA task           |
|-------------------------------------|-------------------|-------------------|
|                                     | Accuracy (%)      | Accuracy (%)      |
| Shrinkage LDA [54]                  | 60.91±9.07        | 75.60±6.69        |
| IDPF [55]                           | 78.56             | 91.15             |
| Linear Fusion [56]                  | 74.67±8.41        | 87.99±6.67        |
| Tensor Fusion [56]                  | 73.10±8.64        | 87.99±5.80        |
| $p$ th-order Polynomial Fusion [56] | 75.90±8.89        | 87.24±6.14        |
| FGANet [11]                         | 78.59±8.86        | 91.96±5.82        |
| GLM+CSP [9]                         | 79.48±11.32       | 92.24             |
| <b>mBGT (ours)</b>                  | <b>80.48±9.26</b> | <b>93.35±5.22</b> |

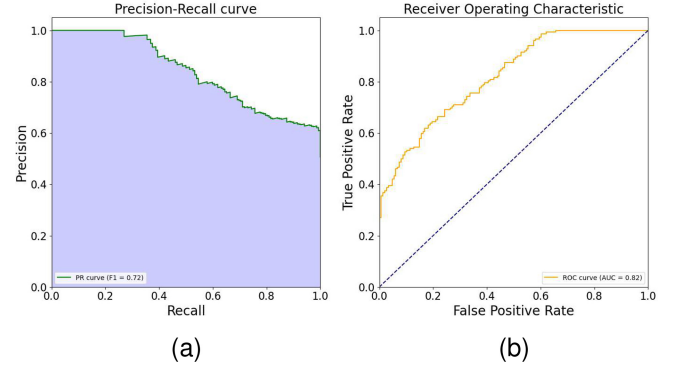


Fig. 4. The results of MI task. (a) shows the PR curve and F1 score, (b) is the ROC curve.

Graph Transformer [42] and Grgahormer [46], in both MA and MI tasks. To demonstrate the effectiveness of our spatiotemporal information enhanced EEG-fNIRS fusion strategy, we train other graph Transformers on (1) EEG data only; (2) fNIRS data only; and (3) simultaneous EEG-fNIRS data.

## B. Results

We compare the proposed model to other deep learning-based EEG-fNIRS signal fusion method in terms of mean accuracy on both MI and MA tasks, where the results of baselines are obtained from their original papers. Table II summarizes the results. In addition, micro F1 score and AUC are provided to evaluate our model on two tasks.

1) *Motor Imagery Task*: As Table II shows, mBGT outperforms other baselines from the perspective of accuracy, achieving 80.48%. In particular, the classification accuracy of mBGT on MI task is about 1-point higher than the second-best model GLM+CSP. This indicates that the effectiveness of mBGT than others in MI task. Note that all models have relatively similar standard deviations. Figure 4 (a) shows the PR (Precision Recall) curve of mBGT, and Figure 4 (b) is the ROC curve of mBGT. As Figure 4 (a) demonstrates, mBGT has high precision  $P$  and high recall  $R$  simultaneously, and the F1 score is 72.67%. In Figure 4 (b), the AUC of mBGT is 82.11%, showing the significant trade-off between TPR and FPR. Additionally, compared to other graph Transformers, mBGT demonstrates significantly higher accuracy, F1 score, and AUC in MI task (see Table III), highlighting its superiority in multimodal brain signal processing. Overall, the classification results prove that mBGT achieves great performance

TABLE III  
COMPARISON BETWEEN MBGT AND OTHER GRAPH  
TRANSFORMER MODELS IN MI TASK

| Method                         | MI task           |                   |                    |
|--------------------------------|-------------------|-------------------|--------------------|
|                                | Accuracy (%)      | F1(%)             | AUC (%)            |
| Graph Transformer (EEG only)   | 62.56±8.06        | 58.93±6.35        | 63.91±6.71         |
| Graph Transformer (fNIRS only) | 64.42±8.71        | 59.24±8.93        | 66.40±7.98         |
| Graph Transformer (EEG+fNIRS)  | 68.54±7.52        | 62.68±7.78        | 70.29±6.77         |
| Graphormer (EEG only)          | 69.83±9.31        | 64.84±6.65        | 71.19±7.32         |
| Graphormer (fNIRS only)        | 67.02±8.71        | 63.41±7.13        | 70.25±8.86         |
| Graphormer (EEG+fNIRS)         | 72.47±7.62        | 66.29±6.69        | 74.44±7.59         |
| <b>mBGT (ours)</b>             | <b>80.48±9.26</b> | <b>72.67±6.05</b> | <b>82.11± 7.43</b> |

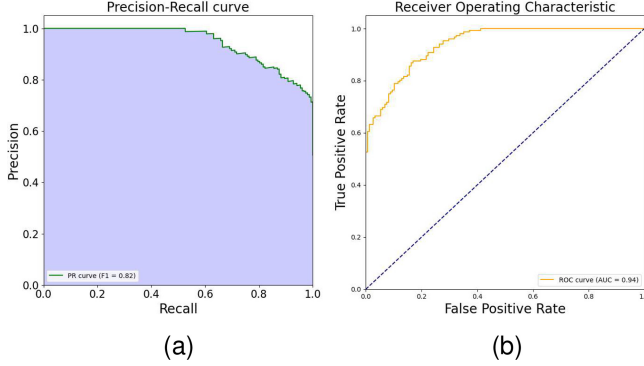


Fig. 5. The results of MA task. (a) is the PR curve and F1 score, (b) indicates the ROC curve.

in MI task. This reveals that capturing the spatiotemporal relations of multimodal brain signals when subjects are doing motor imagery task can significantly explore the information transmission mechanisms among brain regions.

2) *Mental Arithmetic Task*: For MA task, the proposed mBGT achieves the highest classification accuracy with 93.35% (see Table II), showing great improvement over other baselines. Moreover, Fig. 5 (a) is the PR curve of mBGT in MA task, while Fig. 5 (b) demonstrates the ROC curve of mBGT. In Fig. 5, the F1 score of mBGT is 82.64% with high values of both precision and recall, and the AUC of mBGT achieves 94.20%. Similarly, as shown in Table IV, mBGT shows better performance than other graph Transformer models. These results indicate the great performance of our model in MA task, showing the superiority of mBGT in learning the features of multimodal brain signals when subjects are doing mental arithmetic task. In conclusion, the overall results of both MI and MA tasks prove that mBGT can effectively combine the strengths of each modality so as to improve the quality of multimodal representations. Therefore, compared to other models, mBGT has demonstrated its ability to obtain high-quality representations of multimodal signal data, benefiting both downstream tasks.

### C. Ablation Study

To validate the effectiveness of each component in mBGT, we conduct ablation study on both MI and MA datasets. Firstly, we evaluate the impact of the temporal information embedding strategy on model performance. Then, the impact of the spatial embedding component is evaluated.

TABLE IV  
COMPARISON BETWEEN MBGT AND OTHER GRAPH  
TRANSFORMER MODELS IN MA TASK

| Method                         | MA task           |                   |                    |
|--------------------------------|-------------------|-------------------|--------------------|
|                                | Accuracy (%)      | F1(%)             | AUC (%)            |
| Graph Transformer (EEG only)   | 77.82±8.20        | 67.53±7.72        | 78.52±8.09         |
| Graph Transformer (fNIRS only) | 79.43±8.71        | 73.47±7.54        | 82.85 ±9.22        |
| Graph Transformer (EEG+fNIRS)  | 81.29±8.59        | 74.54 ±6.23       | 84.14±8.31         |
| Graphormer (EEG only)          | 80.25±8.31        | 75.57±9.89        | 84.49±7.22         |
| Graphormer (fNIRS only)        | 78.62±8.85        | 74.25±10.62       | 81.32±7.46         |
| Graphormer (EEG+fNIRS)         | 83.32±7.12        | 79.60±8.33        | 86.27±8.74         |
| <b>mBGT (ours)</b>             | <b>93.35±5.22</b> | <b>82.64±7.02</b> | <b>94.20± 8.77</b> |

1) *Impact of the Temporal Information Embedding Strategy*: To show how the temporal features extracted from EEG data benefit the model performance, we conduct an experiment, removing EEG temporal features when obtaining node embeddings. The results are summarized in Table V, wherein mBGT-T means the model without EEG temporal information embeddings. As Table V shows, the performance of mBGT-T is significantly reduced on both MI and MA datasets. Particularly, compared to mBGT, the accuracies of mBGT-T on MI and MA tasks are decreased by around 4 and 5 points, respectively. In addition, Fig. 6 shows the accuracy improvement of mBGT compared to mBGT-T for all subjects. Circles above the diagonal line indicate subjects having improved brain state detection accuracy when using mBGT. The percentage values (73% on both tasks) are the ratios of the number of subjects having improved results out of all subjects.

The overall results indicate that removing the temporal information embedding component can influence the model performance, suggesting that the EEG temporal features are indispensable when obtaining node embeddings. Because EEG signals have high temporal resolution carrying rich time-varying information, integrating EEG temporal information into the node features can capture sufficient dynamic knowledge for high-quality representations. Moreover, combining EEG temporal information with fNIRS features can capture the temporal correlation of multimodal data, therefore, enhancing the ability of the model to explore the complex dynamic information interaction.

2) *Impact of the Spatial Embedding Component*: To verify the effectiveness of the spatial embedding component, we omit the spatial attention matrix  $\mathbf{A}_t^s$  when training ST-Graph Transformer. The results are shown in Table V, wherein mBGT-S is the model without the spatial embedding. Similarly, the performance of mBGT-S is also reduced in both MA and MI datasets compared to mBGT. Specifically, the accuracies of mBGT-S on both the MI and MA tasks are decreased by around 5% and 6%. Fig. 7 shows the accuracy improvement of mBGT compared to mBGT-S for all subjects. The ratios of the number of subjects with improved performances out of all subjects are 70% and 67% on MI task and MA task, respectively. This indicates that the spatial embedding component is also an essential part of mBGT, and capturing spatial information of brain signals is vital for multimodal data representation. Notably, the spatial

TABLE V  
THE RESULTS OF MBGT AND ITS VARIANTS ON MI AND MA TASKS

| Method      | MI task      |              |              | MA task      |              |              |
|-------------|--------------|--------------|--------------|--------------|--------------|--------------|
|             | F1 (%)       | Accuracy (%) | AUC (%)      | F1 (%)       | Accuracy (%) | AUC (%)      |
| mBGT-T      | 68.44        | 76.21        | 77.02        | 78.23        | 88.76        | 89.10        |
| mBGT-S      | 66.32        | 75.07        | 75.50        | 76.31        | 86.92        | 87.11        |
| <b>mBGT</b> | <b>72.67</b> | <b>80.48</b> | <b>82.11</b> | <b>82.64</b> | <b>93.35</b> | <b>94.20</b> |

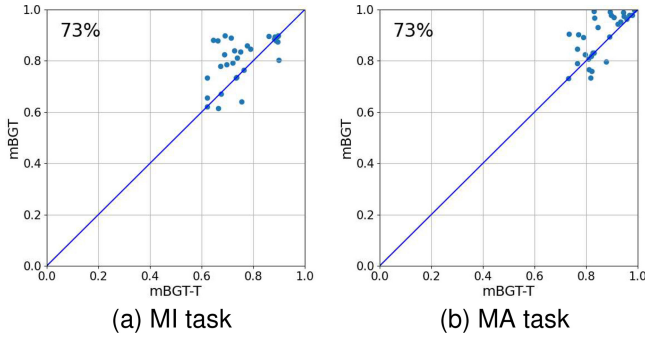


Fig. 6. The accuracy improvement of mBGT compared to mBGT-T for all subjects.

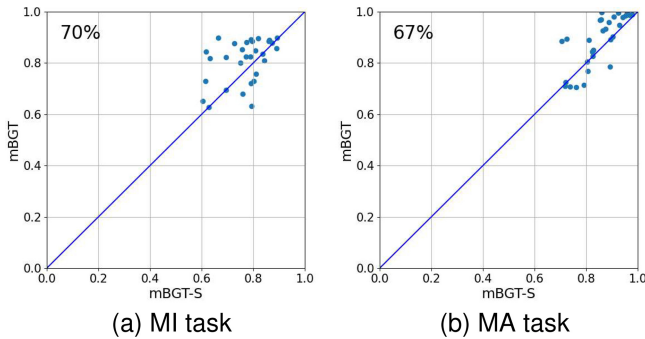


Fig. 7. The accuracy improvement of mBGT compared to mBGT-S for all subjects.

embedding component can encode the spatial correlations between fNIRS signals that have high spatial resolution, extracting more effective spatial information. On the contrary, without the spatial embedding, the multimodal brain signal representations may lack sufficient knowledge and cannot reflect the information interaction mechanism between brain regions. Therefore, if the spatial embedding is removed, the model performance could be reduced.

## VI. CONCLUSION

In this paper, we present a novel multimodal brain graph Transformer model, named mBGT, to generate high-quality multimodal representations of brain signals. Specifically, we first utilize an interpolation-based data alignment method to align the collected EEG and fNIRS signals, ensuring that they are mapped onto the same temporal and spatial space. Then, we construct a dynamic brain graph based on the aligned multimodal EEG-fNIRS data to capture the spatiotemporal relations of brain signals. Specifically, we take advantage of the high temporal resolution of EEG and superior spatial

resolution of fNIRS to enhance node features and calculate edge strength, respectively. Subsequently, we design a spatiotemporal information enhanced graph Transformer (ST-Graph Transformer) to learn brain signal representations over the constructed dynamic brain graph, fusing the multimodal information effectively. As such, our proposed method can effectively capture the spatiotemporal relations of EEG-fNIRS signals and fully utilize the strengths of each modality. Our model is evaluated on two downstream tasks: mental arithmetic (MA) and motor imagery (MI). The experimental results demonstrate the superiority of mBGT over strong baselines.

Although our model has been evaluated on a reliable benchmark dataset, which contains two crucial brain activity detection tasks, it would be more robust to evaluate our model on more brain disease-related datasets, such as epilepsy datasets. Therefore, in future work, we will extend our model to other brain health data, enhancing the generalization ability of our model and applying it to various brain disease detection tasks. In addition, because brain signal data is sensitive to noises, and some artefact noises are highly similar to the real signal data, denoising is still a challenging task for signal data processing. Our future work will also pay attention to signal data augmentation, including denoising and missing signal completion.

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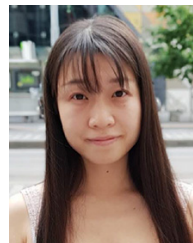
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